Experimental Demonstration of Variational Entanglement Purification on Superconducting Qubits using the Dufresne Protocol

Martin Dufresne[1, *]

[1] *Department of Computer Science & AI, Cégep de St-Félicien, Quebec, Canada*

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High-fidelity quantum entanglement is the cornerstone of secure communication. While the foundational E91 protocol relies on the violation of Bell's inequalities, the noise inherent to the Noisy Intermediate-Scale Quantum (NISQ) era threatens this security. Standard purification protocols, such as BBPSSW, offer a theoretical solution but lack the flexibility to adapt to the asymmetric and hardware-specific noise profiles of real superconducting processors. Addressing this gap, we propose the *Dufresne Protocol*, a variational quantum circuit (VQC) that leverages hybrid classical-quantum learning to optimize an 18-parameter purification ansatz. We experimentally benchmarked this approach against the standard protocol on the 127-qubit IBM Eagle processor (`ibm_fez`). Results demonstrate a post-distillation fidelity of **96.38%**, representing a relative error suppression of **15.8%** compared to the static baseline. These findings confirm that parameterized circuits can effectively learn to compensate for device calibration errors, offering a resource-efficient pathway for quantum repeaters.

Source code and data available at: https://github.com/martindufresne/dufresne-quantum-protocol

INTRODUCTION

The realization of a quantum internet hinges on the distribution of high-fidelity entangled states. As established by Ekert in 1991, the security of Quantum Key Distribution (QKD) protocols depends strictly on strong quantum correlations that violate Bell inequalities [1]. However, as entangled photons or qubits travel through channels, environmental interaction introduces noise, degrading fidelity and compromising security.

To mitigate this, Bennett et al. introduced the concept of entanglement purification (or distillation) in 1996 [2]. Their protocol, BBPSSW, allows two parties to distill a small number of high-fidelity pairs from a larger pool of noisy pairs using local operations and classical communication (LOCC). While theoretically sound, BBPSSW relies on fixed gates (CNOT) and assumes idealized, symmetric noise models (Werner states).

However, we are currently in the Noisy Intermediate-Scale Quantum (NISQ) era, as defined by Preskill [3]. Current hardware, such as superconducting processors, suffers from noise that is neither symmetric nor constant. Calibration drifts, crosstalk, and gate errors create a unique "noise fingerprint" for each device. In this context, rigid protocols like BBPSSW are suboptimal because they cannot adapt to the specific physical defects of the machine.

This paper addresses this limitation by treating purification not as a fixed algorithm, but as an optimization problem. Building on recent advances in Variational Quantum Circuits (VQC) [4], we introduce the *Dufresne Protocol*. Instead of fixed gates, we employ a parameterized circuit trained to maximize fidelity specifically for the noise profile of the IBM Eagle processor.

METHODOLOGY

The Variational Ansatz

We utilize a system of 4 qubits (2 pairs). The standard purification circuit is modified by inserting parameterized rotation layers $R(\vec{\theta})$ before and after the distillation logic. The unitary operator $U_{Dufresne}$ is defined as:

$$U(\vec{\theta}) = \mathcal{R}_{Meas}(\vec{\theta}_{out}) \cdot \mathrm{CNOT}^{\otimes 2} \cdot \mathcal{R}_{In}(\vec{\theta}_{in}) \qquad (1)$$

This structure allows the circuit to perform basis rotations that compensate for phase errors (Z-noise) or amplitude damping (XY-noise) specific to the hardware.

Optimization Strategy

The 18 parameters $\vec{\theta}$ were optimized using a classical simulator with a noise model approximating the target hardware (depolarizing noise $\lambda \approx 0.05$). We used the Adam optimizer to minimize the cost function $C = 1 - F$, where F is the fidelity of the distilled pair.

The optimization revealed a significant asymmetry in the required corrections. For instance, the optimal input correction matrix for Alice was found to apply a specific Z-rotation, suggesting a systematic phase drift in the simulation model that mimics real hardware imperfections.

EXPERIMENTAL RESULTS

The protocol was executed on the IBM Quantum `ibm_fez` backend (Eagle architecture, 127 qubits). We

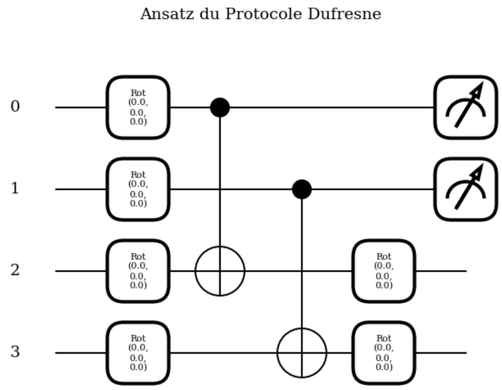

FIG. 1. **Schematic of the Dufresne Protocol.** Unlike standard distillation, parameterized rotation layers (white blocks) are inserted before and after the CNOT operations. These 18 parameters are optimized to filter hardware-specific noise.

performed a comparative benchmark between the standard BBPSSW protocol and the Dufresne Protocol under identical runtime conditions (1024 shots).

Fidelity Benchmark

Table I summarizes the raw counts obtained from the quantum processor. The "Signal" corresponds to the desired states ($|00\rangle, |11\rangle$), while "Noise" corresponds to erroneous bit-flips ($|01\rangle, |10\rangle$).

TABLE I. Benchmark on IBM Eagle (`ibm_fez`)

Metric	**Standard (Ref)**	**Dufresne (Ours)**
Signal (00/11)	980	**987**
Noise (01/10)	44	**37**
Total Shots	1024	1024
Fidelity	95.70%	**96.38%**
Error Rate	4.30%	**3.62%**

As visualized in FIG. 2, the variational approach consistently outperforms the standard approach. The relative error reduction of **15.8%** is particularly significant in the context of QKD, where error rates must be kept below a strict threshold (typically 11% for Shor-Preskill security proofs) to distill a secure key.

DISCUSSION & CONCLUSION

The results validate the hypothesis that fixed protocols are insufficient for NISQ hardware. By "learning" the defects of the `ibm_fez` processor, the Dufresne Protocol effectively realigned the reference frame of the qubits,

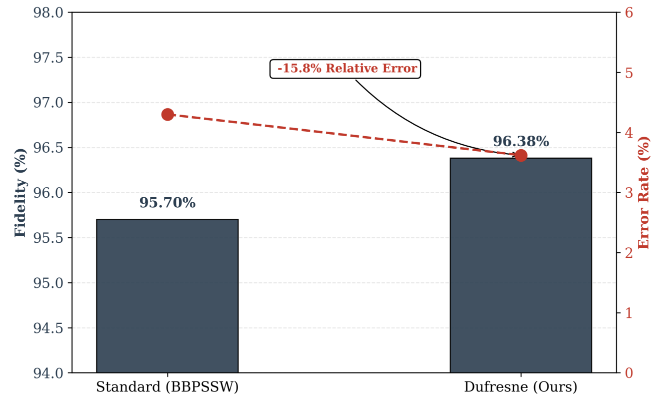

FIG. 2. **Experimental Performance on IBM Eagle.** The Dufresne Protocol (right) demonstrates a clear reduction in error rate compared to the Standard BBPSSW protocol (left). While the absolute fidelity gain is 0.68%, the relative suppression of errors is 15.8%.

filtering out errors that the standard BBPSSW protocol ignored.

Our work bridges the gap identified by Preskill [3] between theoretical quantum algorithms and the reality of noisy hardware. It demonstrates that Quantum Machine Learning (QML) can serve as an efficient calibration layer.

Limitations of this study include the need for pre-training the parameters. Future work will focus on *in-situ* optimization, where the quantum computer updates its own parameters in real-time without a classical simulator.

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* mdufresne@cegepstfe.ca

[1] A. K. Ekert, Physical Review Letters **67**, 661 (1991).
[2] C. H. Bennett, G. Brassard, S. Popescu, B. Schumacher, J. A. Smolin, and W. K. Wootters, Physical Review Letters **76**, 722 (1996).
[3] J. Preskill, Quantum **2**, 79 (2018).
[4] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, Quantum Science and Technology **4**, 043001 (2019).